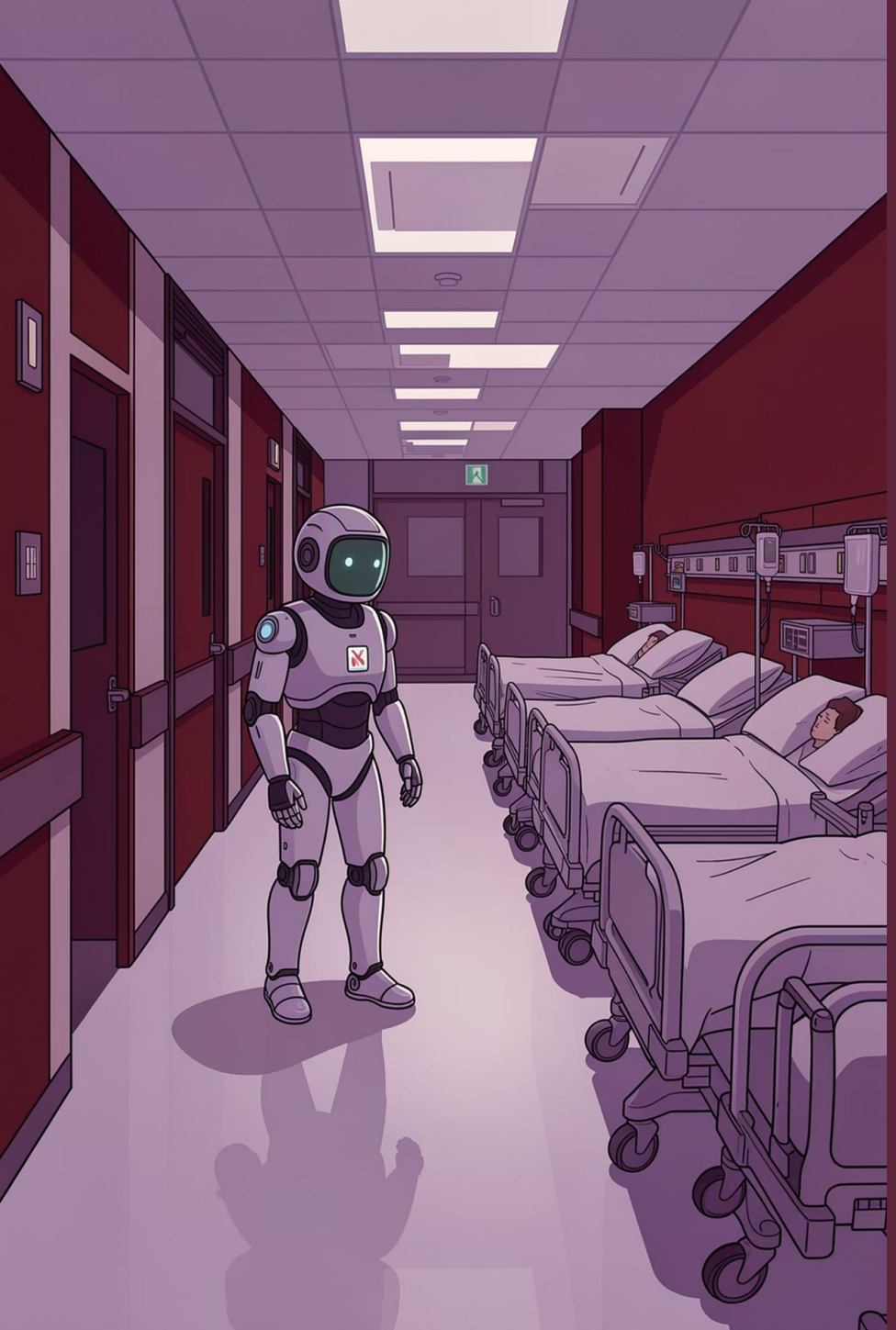


Hospital Health Monitoring Robot using IoT

HNDCSAI – Robotics & IoT

Group members: Piratheesha, Fathima

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Introduction

A smart IoT-based robotic system for automated patient monitoring.

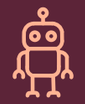
- **The problem**
  - Manual vital checks are time-consuming
  - Prone to human error
  - Limited continuous monitoring in hospitals

- **Our solution**

  A mobile robot that autonomously visits beds, measures vital signs (heart rate, SpO_2, temperature), and uploads real-time data to the cloud using IoT.

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Core Functions



Navigation

Line-following using IR sensor array with predefined patient stop points, with optional manual control



Vital Measurements

Heart rate (BPM), SpO_2 (%), and temperature (°C) measured at each patient location.



Safety & Alerts

Ultrasonic sensor detects obstacles; abnormal readings trigger real-time alerts.

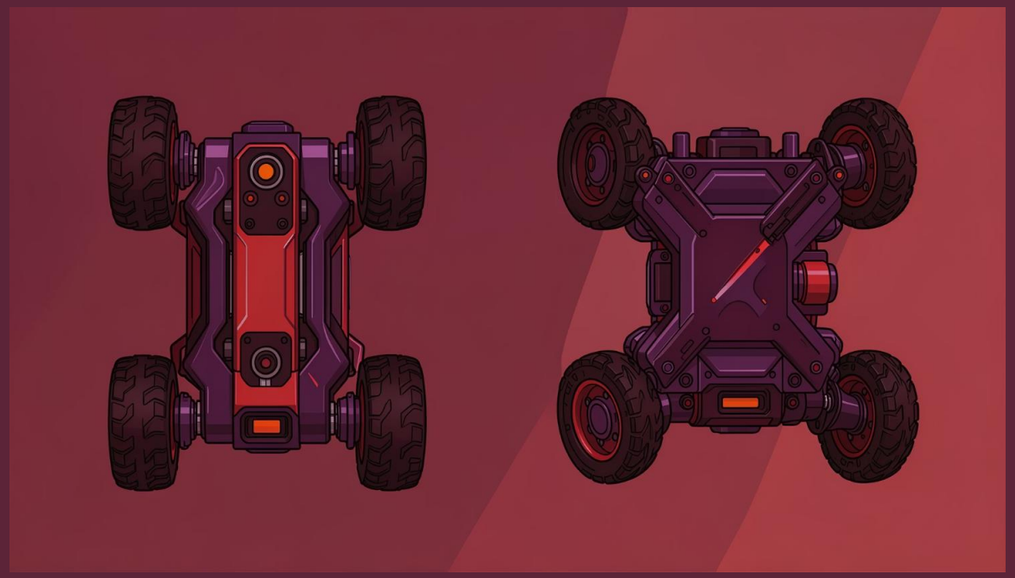


Data Handling

Sensor data is transmitted to Firebase for real-time monitoring and storage.

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Detailed Design — Structure & Sensor Placement



Top Layer (Upper Deck)

- Compact chassis for efficient navigation in hospital ward aisles

Sensor placement

- IR sensor array mounted underneath the front for line following
- MAX30102 (heart rate & SpO_2) and DHT22 temperature sensor mounted on an adjustable arm near the patient
- Ultrasonic sensor positioned at the front for obstacle detection

Bottom Layer (Lower Deck)

- 4x DC gear motors at corners, L298N motor driver at centre, 18650 battery pack at rear, main power switch at edge

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Key Components

ESP32

Central MCU with Wi-Fi for IoT connectivity

L298N

Motor driver for 4 DC motors

4 × DC motors & chassis

4-wheel drive for stable movement

IR sensor array

Line following

MAX30100/30102

Heart rate & SpO2 sensor

DHT22 / MLX90614

Temperature sensing (contact or contactless)

HC-SR04

Ultrasonic obstacle detection

Li-ion batteries

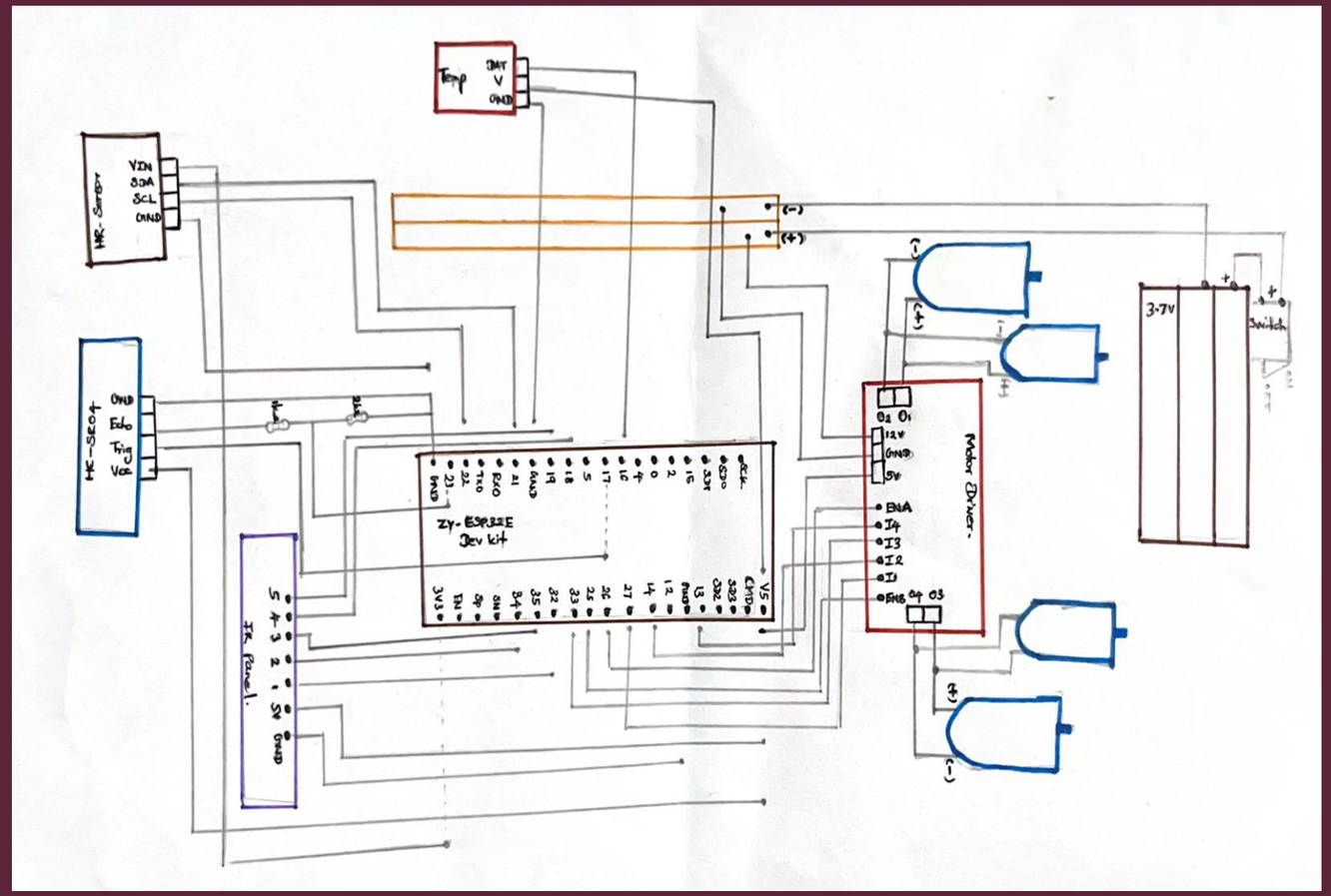
3 × 3.7V cells with power management

Switch & wiring

Power control and interconnections

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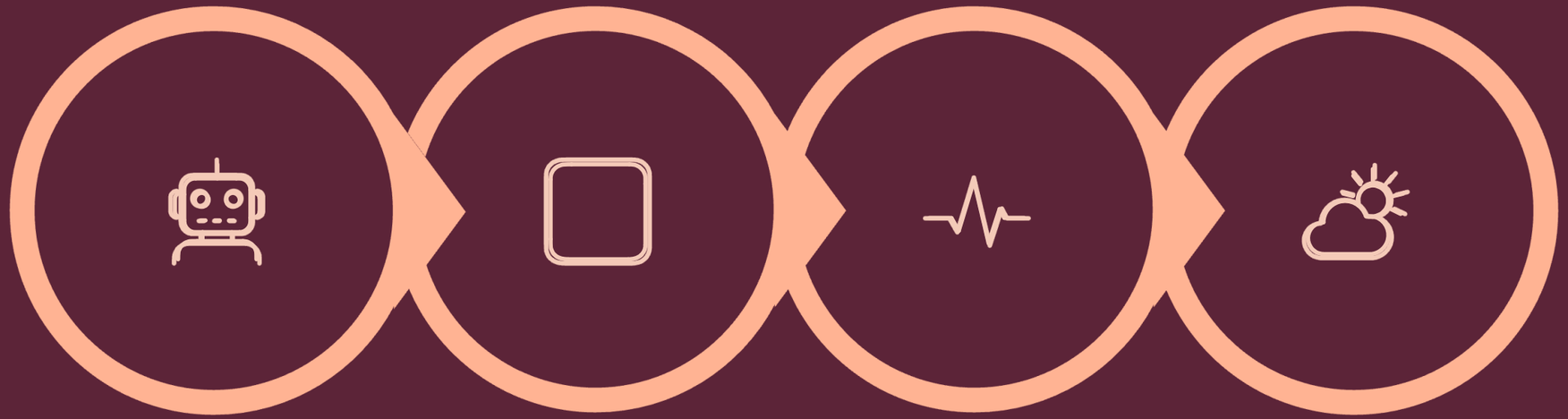
Connections Overview



- ESP32 acts as the central controller connecting all components
- IR sensor array connected to GPIO pins for line following
- Ultrasonic sensor uses TRIG and ECHO pins for distance measurement
- MAX30102 (heart rate & SpO_2) and DHT22 connected via I2C and digital pins
- L298N motor driver connected to control pins for motor movement
- Battery pack supplies power through a switch
- All components share a common ground for stable operation

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Working Process (Runtime Flow)



Follow Line

Stop at Bed

Read Sensors

Upload & Alert

Robot follows the line, identifies bed points, takes measurements, uploads readings to the database and triggers alerts when thresholds are exceeded.

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Conclusion & Demonstration Plan

Conclusion

Prototype demonstrates an affordable, practical approach to semi-automated vital monitoring that can augment nursing workflows and improve patient safety.

Future improvements

Integrate AI for anomaly detection, a mobile app dashboard, and autonomous localisation to remove line dependence.

2-minute demo

We will show: robot following a line, stopping at a mock bed, real-time sensor readings streaming to the cloud, and an alert triggered by an abnormal value.

